

TEMPEST: Soil Gas Well

April 2025 Samples

2026-09-11

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##Run Information

```
cat("Run Information: Brenden Blakley ") #lets you know what section you're in
```

Run Information: Brenden Blakley

```
#set the run date & user name
```

```
sample_year <- "2025"
```

```
sample_month <- "April"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month (.txt file(s))
```

```
GHG_files <- c("Raw Data/20260826_TMP2025_April.txt")
```

```
#Metadata file path
```

```
Metadata_path = "Metadata/TEMPEST_SGAS_202504_ALL_WORKING.csv"
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
log_path <- "GCRew_CH4_QAQC_Log.csv"
```

```
check_log_path <- "GCRew_Check_QAQC_Log.csv"
```

```
# Define final path for data be sure to change year and month and run number
```

```
final_path <- "Processed Data/TMP2025_April.csv"
```

```
##Standard Type ("Actual" or "Half")
```

```
#LTREB, SMARTX, and GENX PW Samples use "Half", others should use "Actual"
```

```
Standard_Type <- "Actual"
```

```
#Did Some Samples require manual area retrieval?
```

```
##List Samples and their areas for each analyte ("CH4", "CO2", or "N2O")
```

```
Manual_Areas <- data.frame(
```

```
  Sample = c("NA"),
```

```
  Area = c("NA"),
```

```
  Analyte = c("NA")
```

```
)
```

```
#Gas in Water Calculations
```

```
#Are All Samples Gas in Water? ("YES" or "NO")
```

```
GIW <- "NO"
```

```
#Are some samples Gas in Water?
```

```
##List Sample IDs
```

```
GIW_Samples <- c()
```

```
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
```

```
TempC = 25 #Equilibration Temperature (Celsius)
```

```
volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
```

```
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g)
```

```
#record any notes about the run or anything other info here:
```

```

run_notes <- ""

#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)

##Set Up
##Read in metadata and create similar sample IDs for matching to samples
##Read In Data Files
##Get Location of Each Sample in Main Data Frame
##Turn Each Sample into it's own Data Frame
##Get Only Area Tables and Retrieve Info
##QAQC Peak Count and Retention Time

## [1] "If there are too many peaks there is likely a problem with the sample or run"

## [1] "the chromatogram should be checked, if there are too few they may be low/blanks"

## [1] "verify that concentration is low or get area manually"

## [1] "FID: Blank1; 1"      "FID: Blank2; 1"      "FID: Oppm; 1"
## [4] "FID: Std_10K_3mL; 1" "FID: Std_10K_6mL; 1" "FID: Std_10K_12mL; 1"
## [7] "FID: Std_10K_24mL; 1" "FID: Blank3; 1"      "FID: Blank4; 1"
## [10] "FID: Blank5; 1"       "FID: Blank6; 1"      "FID: Blank7; 1"
## [13] "FID: Blank8; 1"       "FID: Blank9; 1"      "FID: Blank9; 1"

## [1] "ECD: Blank1; 0"      "ECD: Blank2; 0"      "ECD: Oppm; 0"
## [4] "ECD: Std_10K_3mL; 0" "ECD: Std_10K_6mL; 0" "ECD: Std_10K_12mL; 0"
## [7] "ECD: Std_10K_24mL; 0" "ECD: Blank3; 0"      "ECD: Blank4; 0"
## [10] "ECD: 2025_April_8; 2" "ECD: Blank5; 0"      "ECD: Blank6; 0"
## [13] "ECD: Blank7; 0"       "ECD: Blank8; 0"      "ECD: Blank9; 0"
## [16] "ECD: Blank9; 0"

## [1] "If RTime is wrong check that correct number of peaks were found"

## [1] "if not a peak may have been mislabeled"

## [1] "if peak number is correct then there may be a problem with the run,"

## [1] "may need to get area manually"

## [1] "CH4:  "

## [1] "CO2:  "

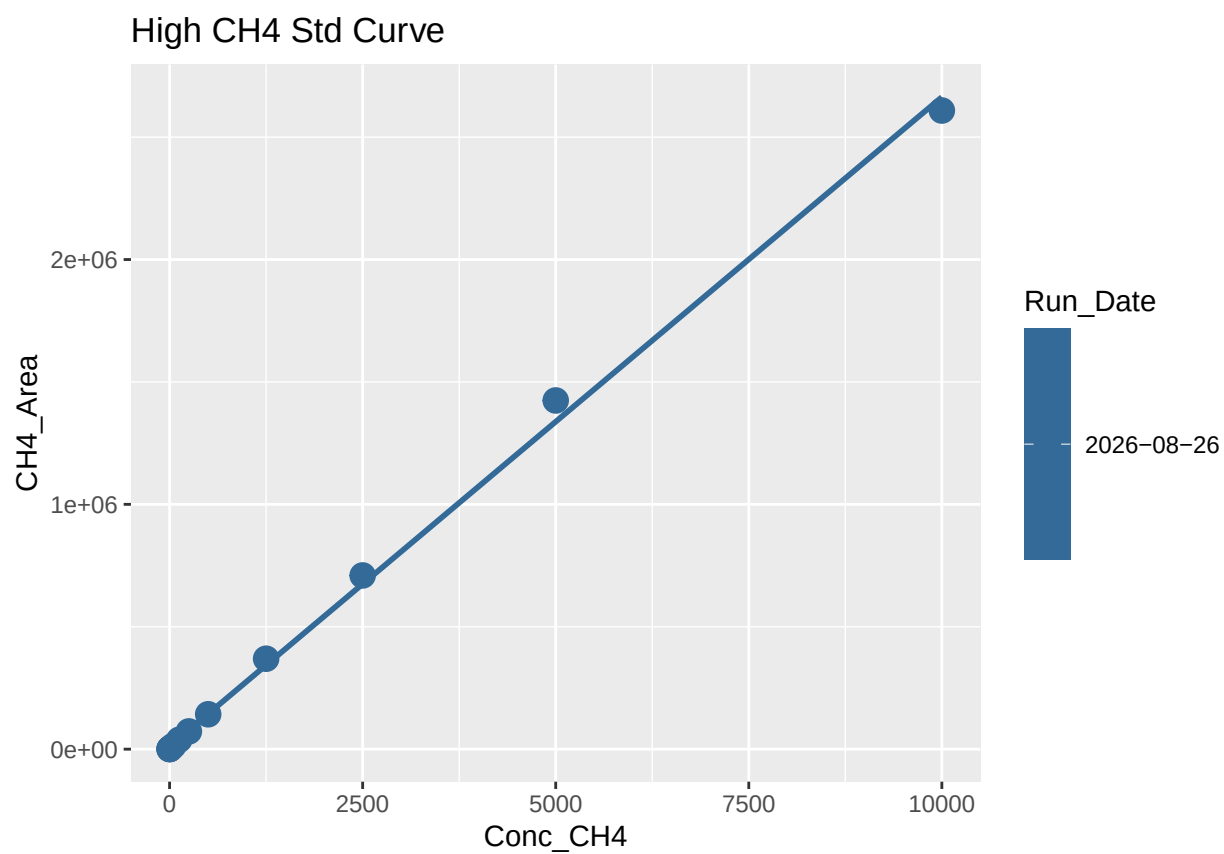
## [1] "N2O:  2025_April_8; 2.755"

##Format Data
##Add Standard Concentrations
##Finalize Data

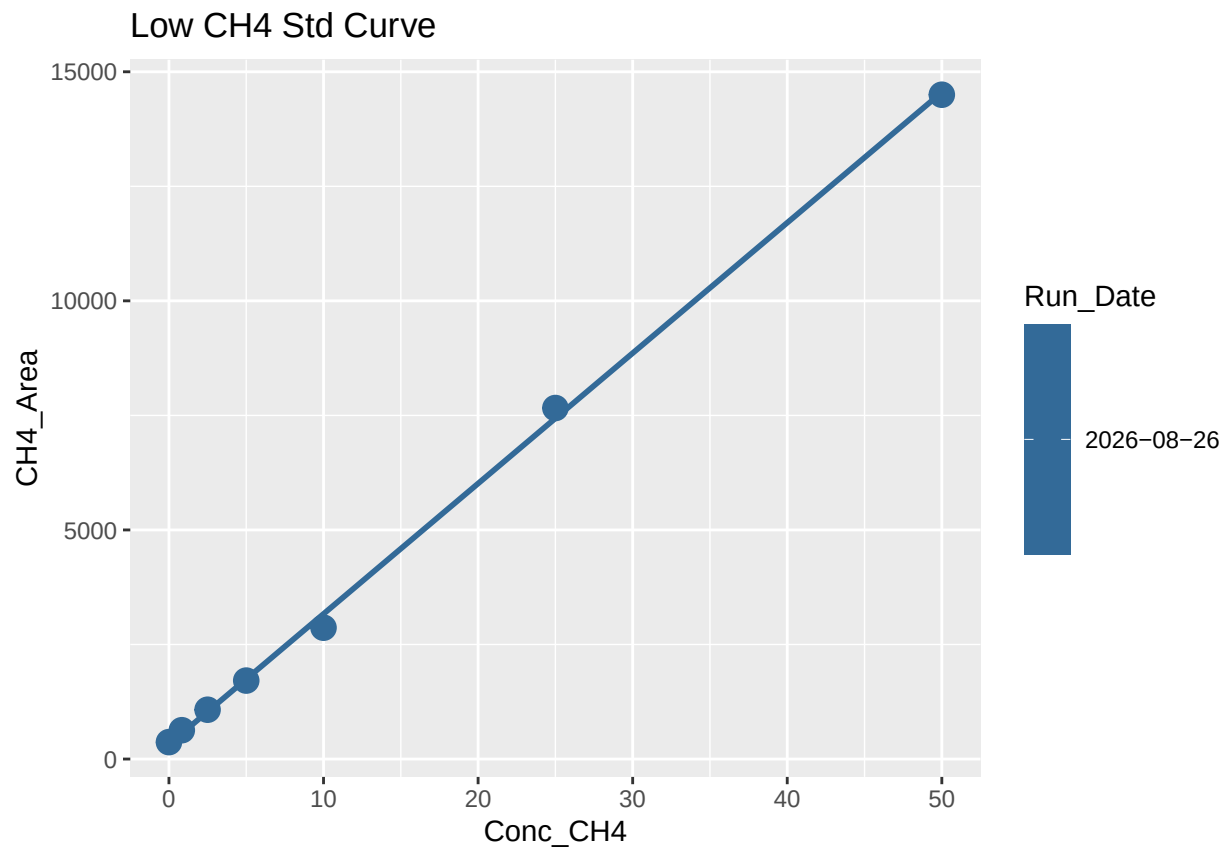
```

1 Assess standard curves CH4

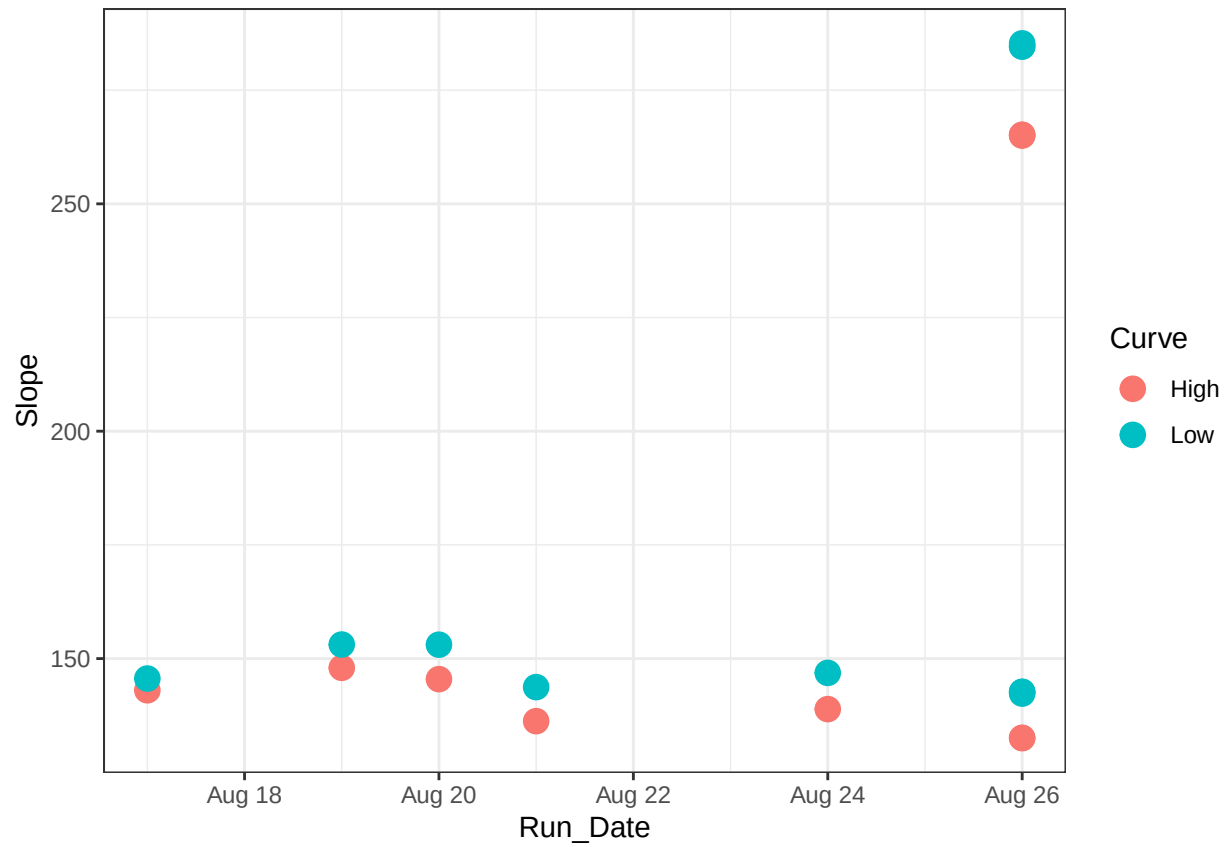
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```



##	Curve	R2	Slope	Intercept	Run_Date	Analyte
## 1	High	0.9994458	136.2406	25980.5484	8/21/2026	CH4
## 2	Low	0.9995311	143.7214	801.9223	8/21/2026	CH4
## 3	High	0.9999168	147.9666	17873.3851	8/19/2026	CH4
## 4	Low	0.9997966	153.0888	407.2899	8/19/2026	CH4
## 5	High	0.9999646	143.0303	7587.2742	8/17/2026	CH4
## 6	Low	0.9996977	145.6125	745.1636	8/17/2026	CH4

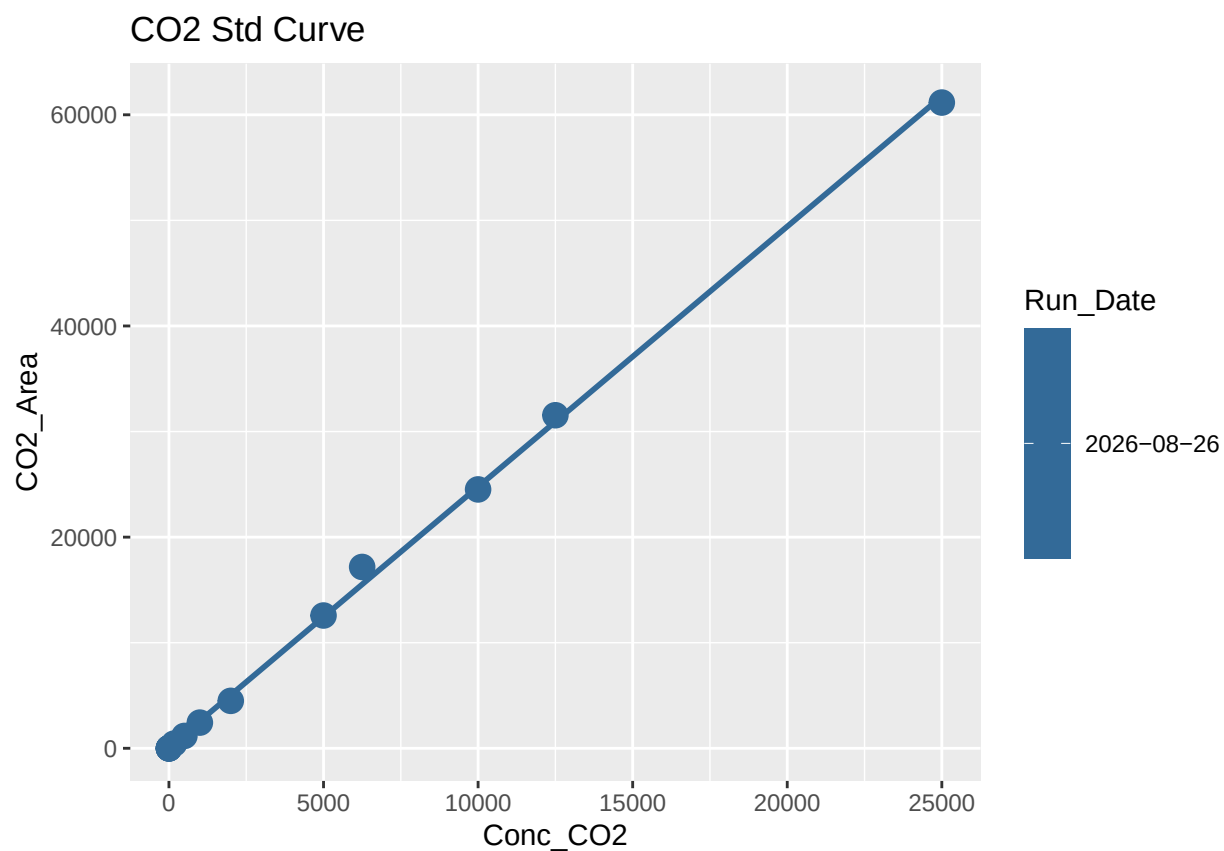


```
## [1] "High_R2 is Good-2026-08-26"
```

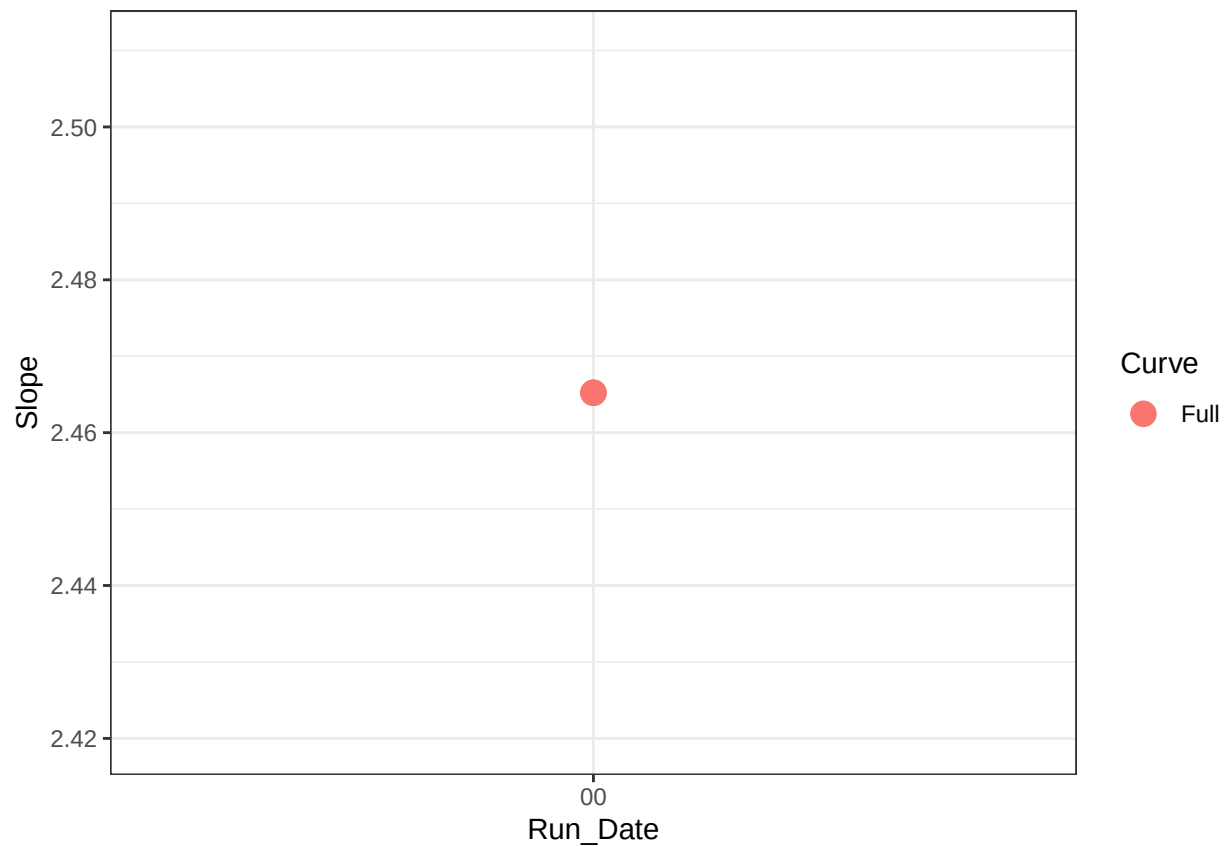
```
## [1] "Low_R2 is Good-2026-08-26"
```

2 Assess standard curves CO2

```
## 'geom_smooth()' using formula = 'y ~ x'
```



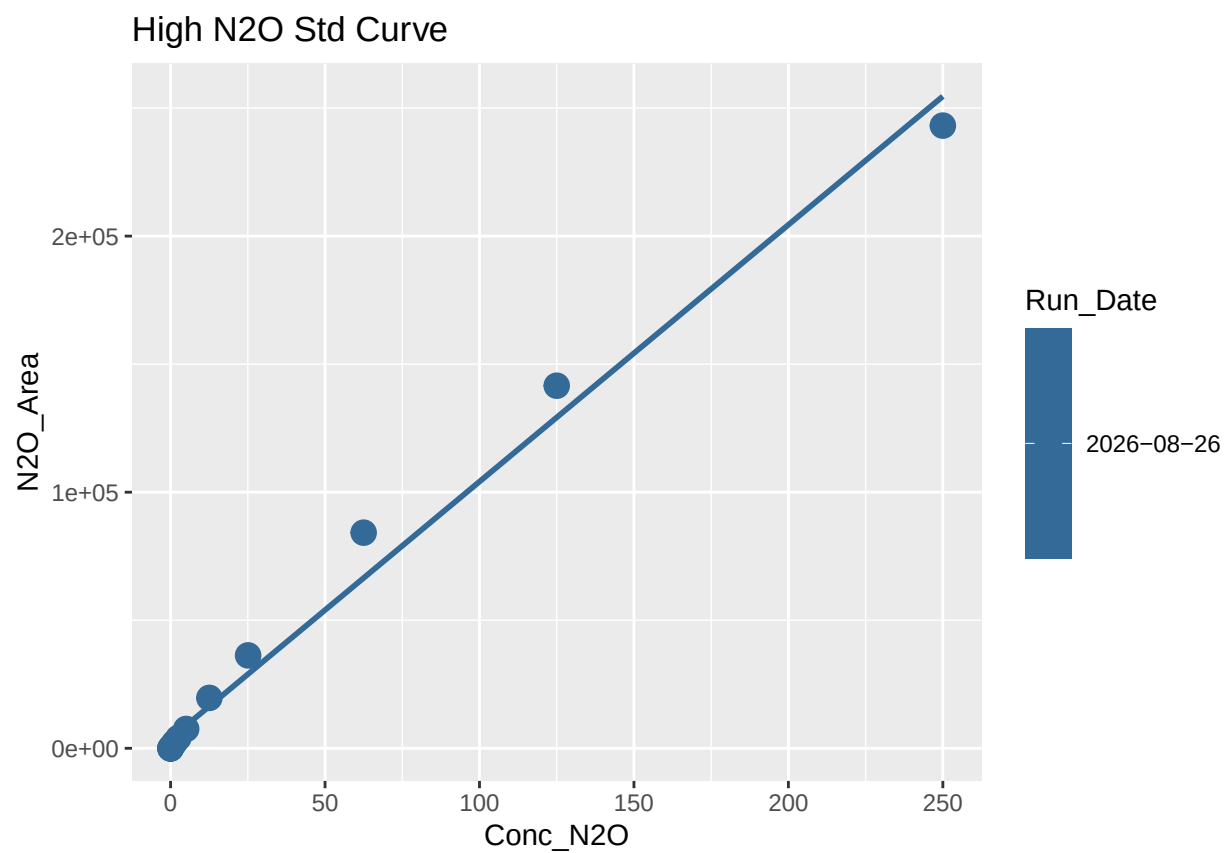
##	Curve	R2	Slope	Intercept	Run_Date	Analyte
## 1	High	0.9994458	136.2406	25980.5484	8/21/2026	CH4
## 2	Low	0.9995311	143.7214	801.9223	8/21/2026	CH4
## 3	High	0.9999168	147.9666	17873.3851	8/19/2026	CH4
## 4	Low	0.9997966	153.0888	407.2899	8/19/2026	CH4
## 5	High	0.9999646	143.0303	7587.2742	8/17/2026	CH4
## 6	Low	0.9996977	145.6125	745.1636	8/17/2026	CH4



```
## [1] "C02_R2 is Good-2026-08-26"
```

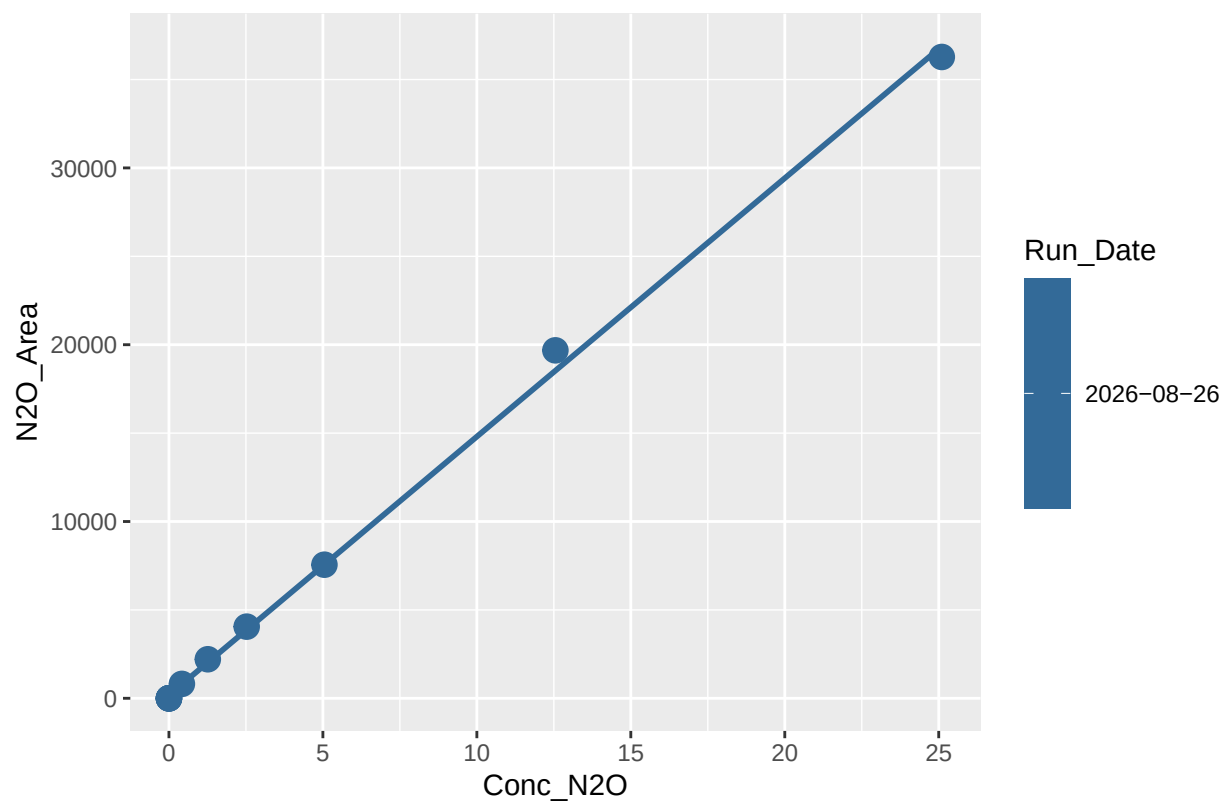
3 Assess standard curves N2O

```
## 'geom_smooth()' using formula = 'y ~ x'
```

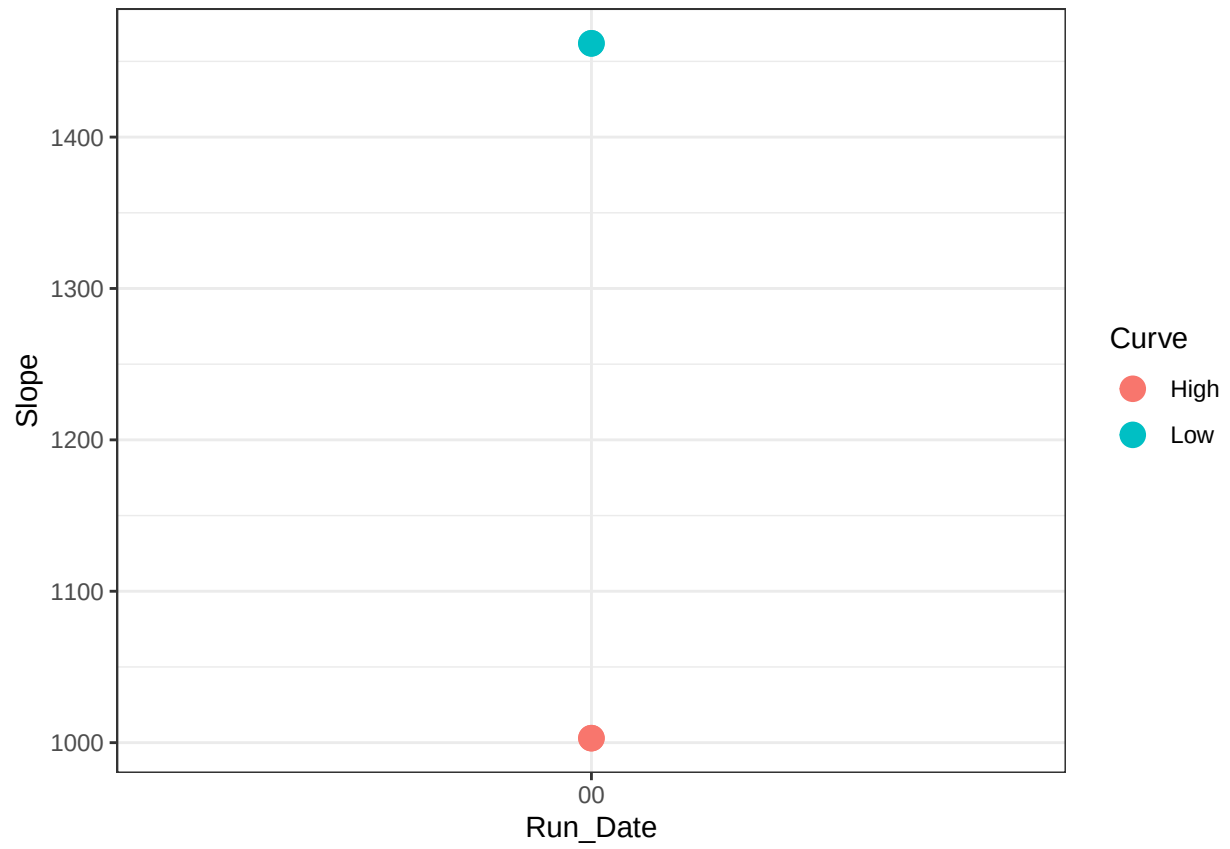


```
## 'geom_smooth()' using formula = 'y ~ x'
```

Low N2O Std Curve



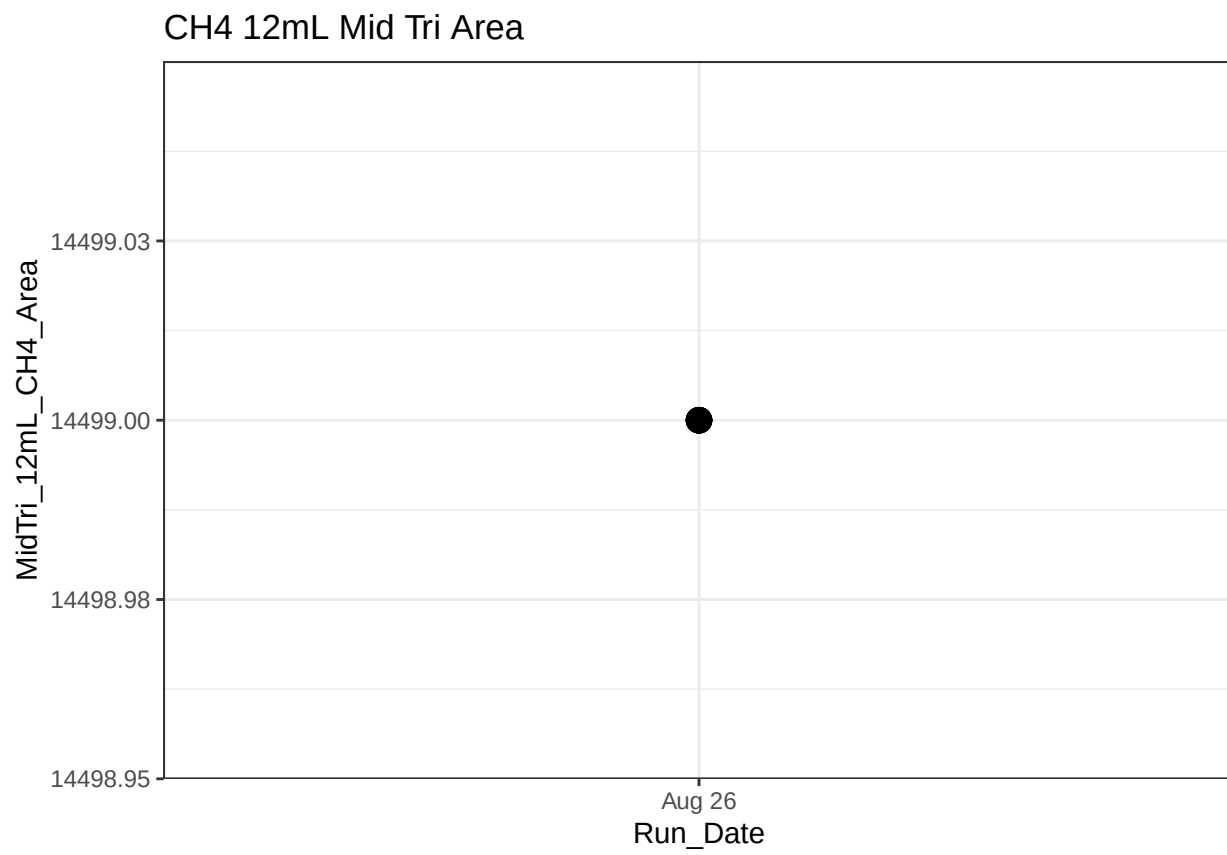
##	Curve	R2	Slope	Intercept	Run_Date	Analyte
## 1	High	0.9994458	136.2406	25980.5484	8/21/2026	CH4
## 2	Low	0.9995311	143.7214	801.9223	8/21/2026	CH4
## 3	High	0.9999168	147.9666	17873.3851	8/19/2026	CH4
## 4	Low	0.9997966	153.0888	407.2899	8/19/2026	CH4
## 5	High	0.9999646	143.0303	7587.2742	8/17/2026	CH4
## 6	Low	0.9996977	145.6125	745.1636	8/17/2026	CH4

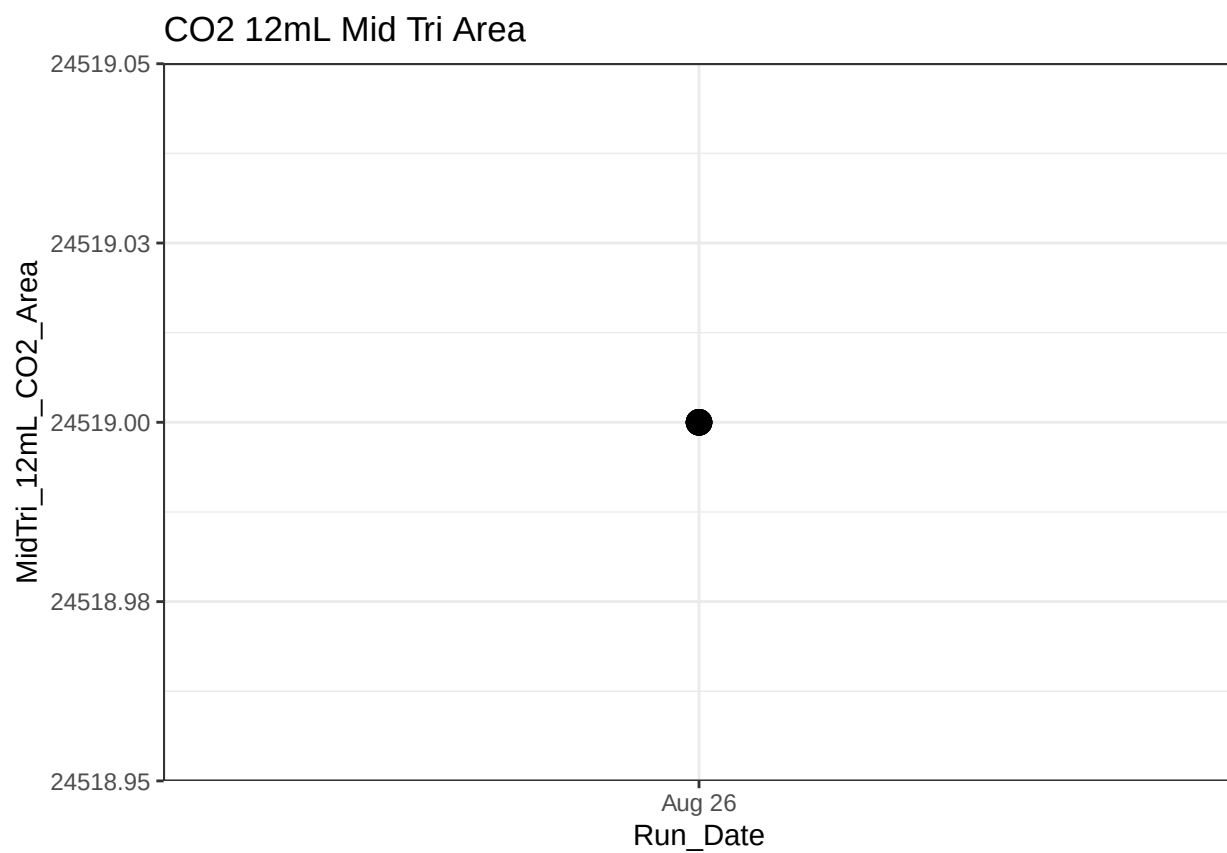


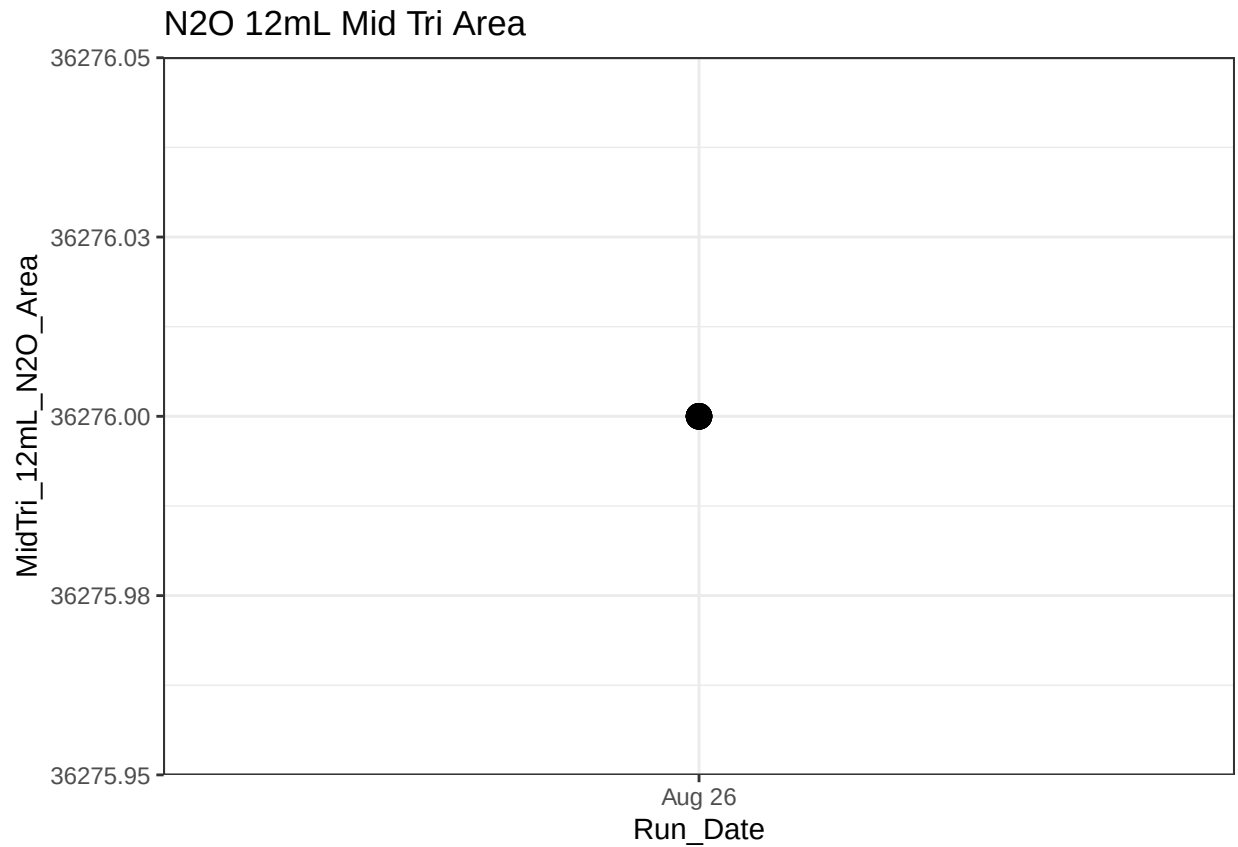
```
## [1] "High_R2 is Bad-2026-08-26"
```

```
## [1] "Low_R2 is Good-2026-08-26"
```

#Plot Check Areas over Time







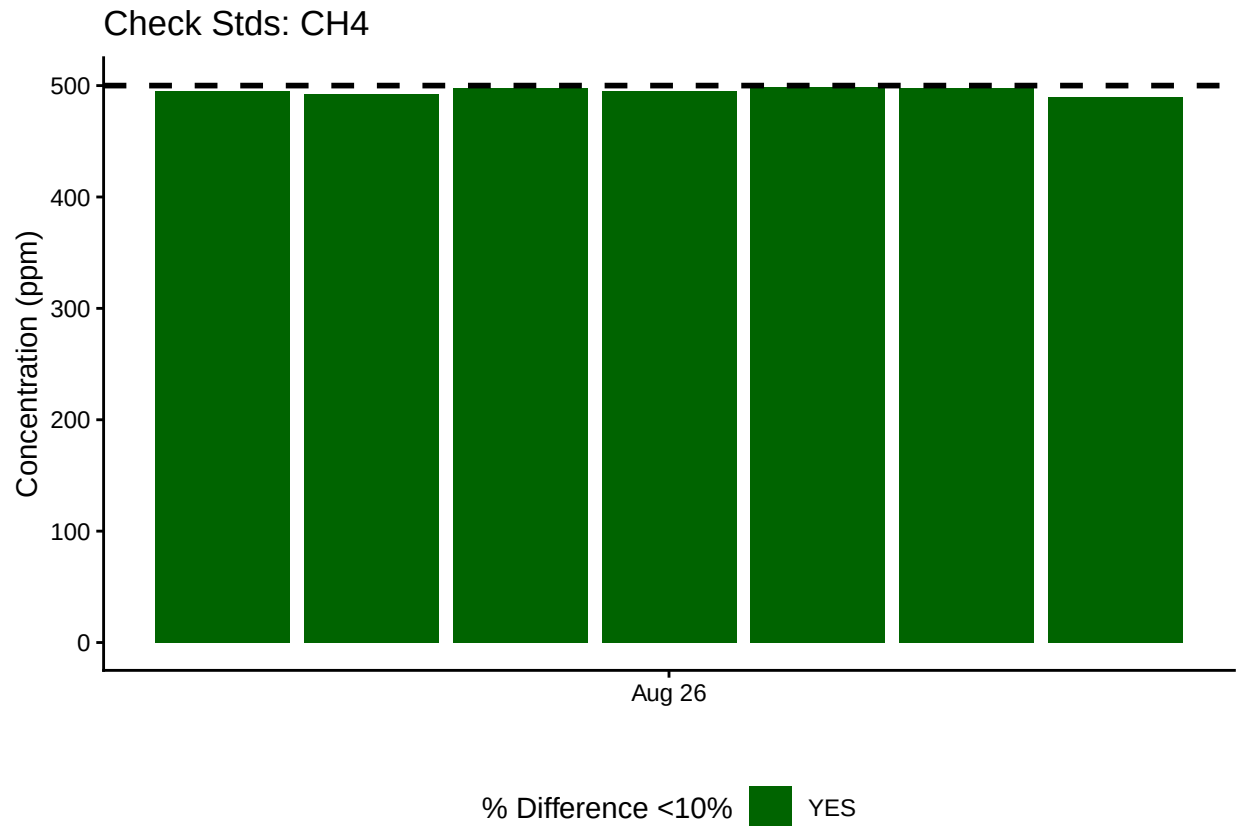
3.1 Now calculate the CH₄ in ppm

3.2 Now calculate the CO₂ in ppm

3.3 Now calculate the N₂O in ppm

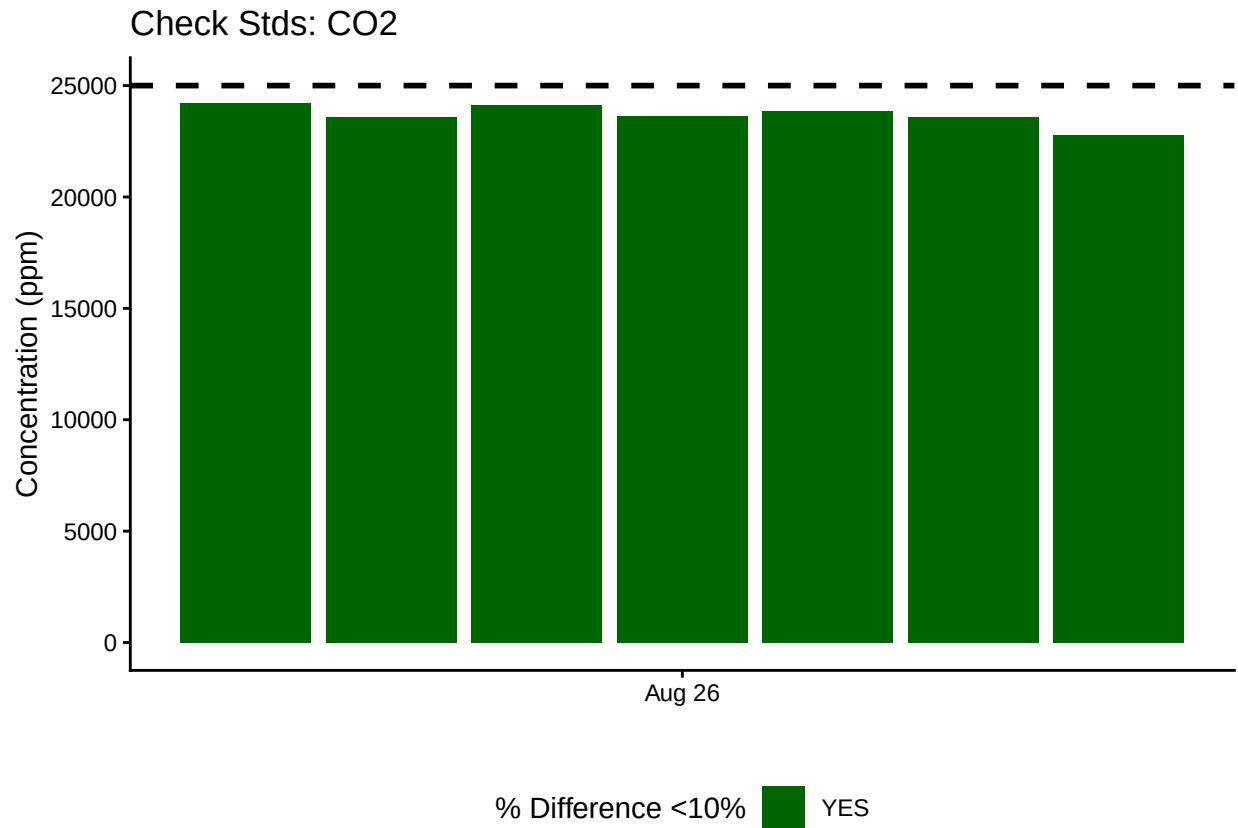
4 Aanlayze the CH₄ Check Standards

>60% of CH₄ Check Standards are within range of expected concentration - PROCEED



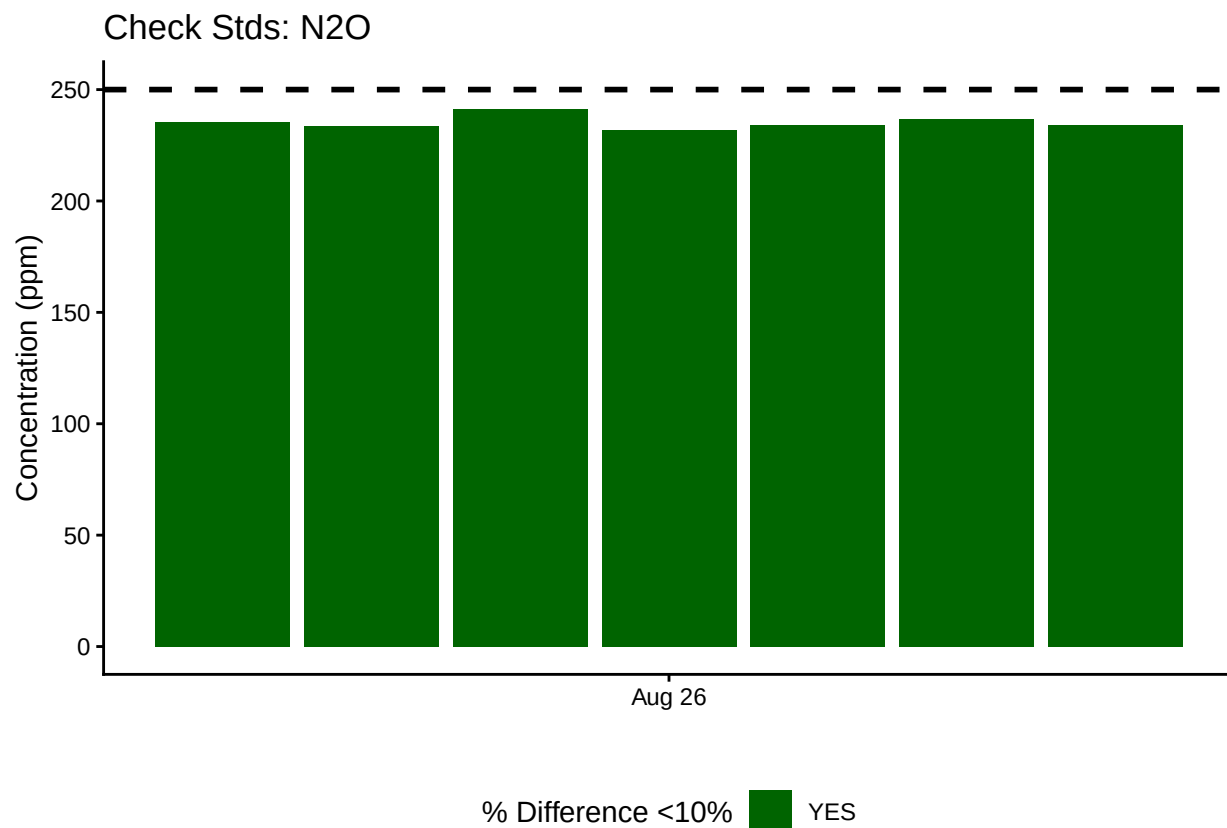
5 Aanlayze the CO2 Check Standards

>60% of CO2 Check Standards are within range of expected concentration - PROCEED



6 Aanlayze the N2O Check Standards

>60% of N2O Check Standards are within range of expected concentration - PROCEED



7 Analyze the CH₄ Blanks

Assess CH₄ Blanks

>60% of CH₄ Blanks are below the lower 25% quartile of samples - PROCEED

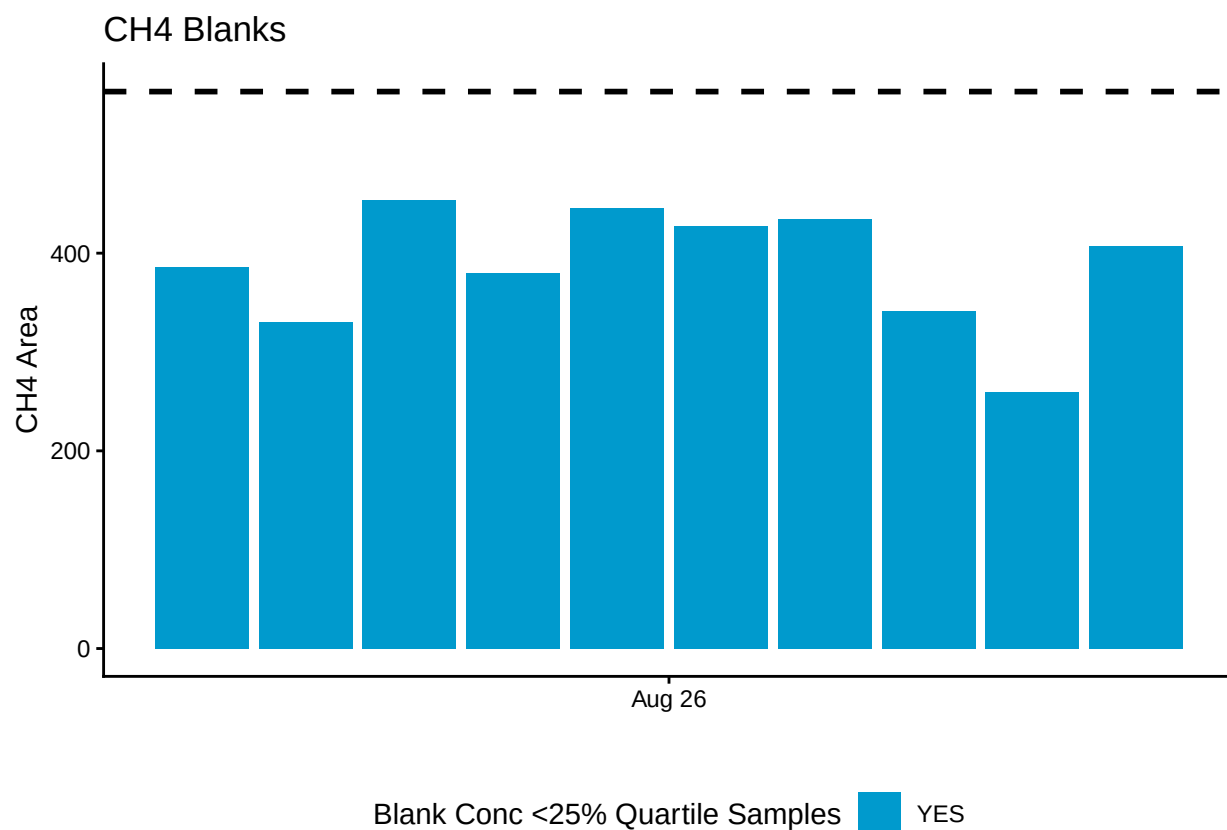


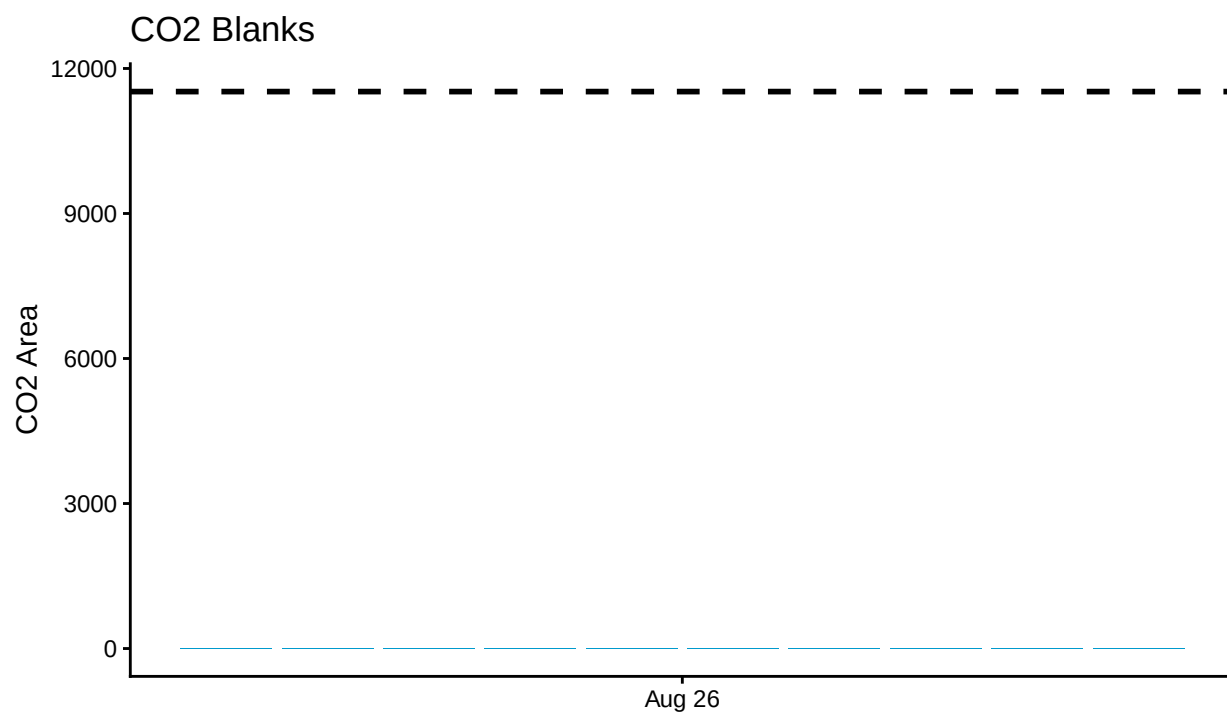
Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	386.6

8 Analyze the CO2 Blanks

Assess CO2 Blanks

>60% of CO2 Blanks are below the lower 25% quartile of samples - PROCEED



Blank Conc <25% Quartile Samples ☒ YES

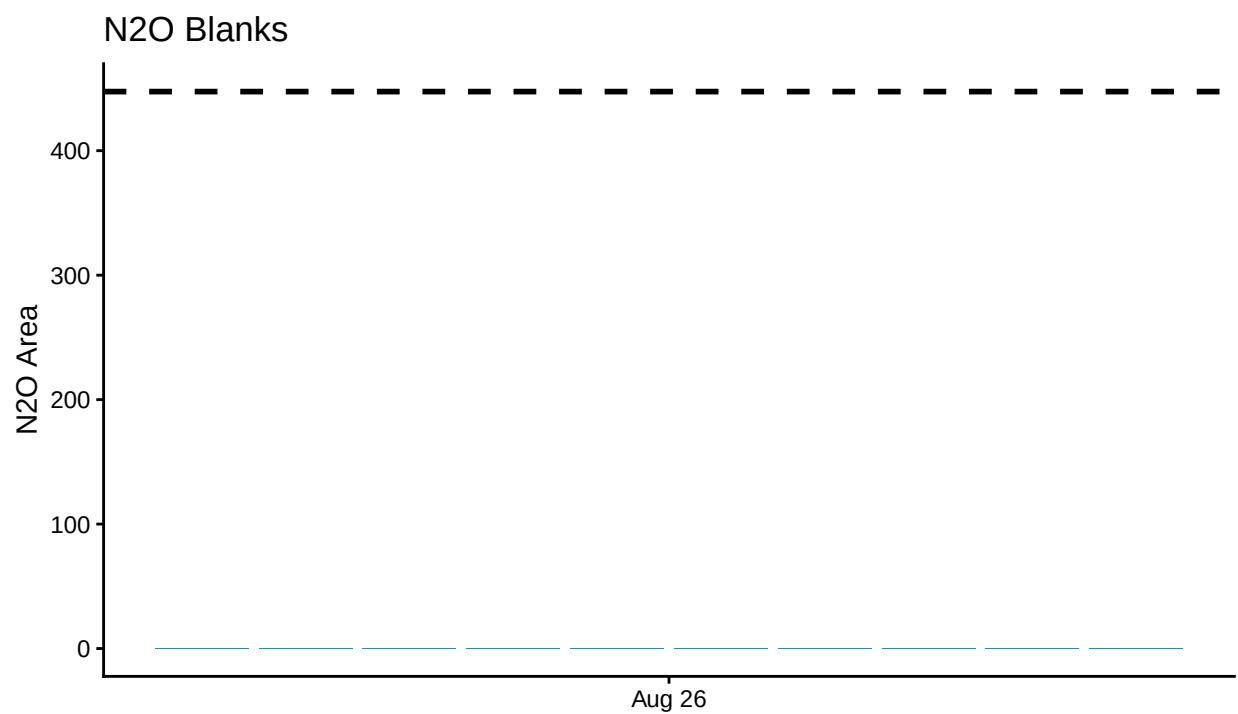
Table 2: Mean Area of Blanks

Test	Blank_Mean_Area
CO2	0

9 Analyze the N2O Blanks

Assess N2O Blanks

>60% of N2O Blanks are below the lower 25% quartile of samples - PROCEED

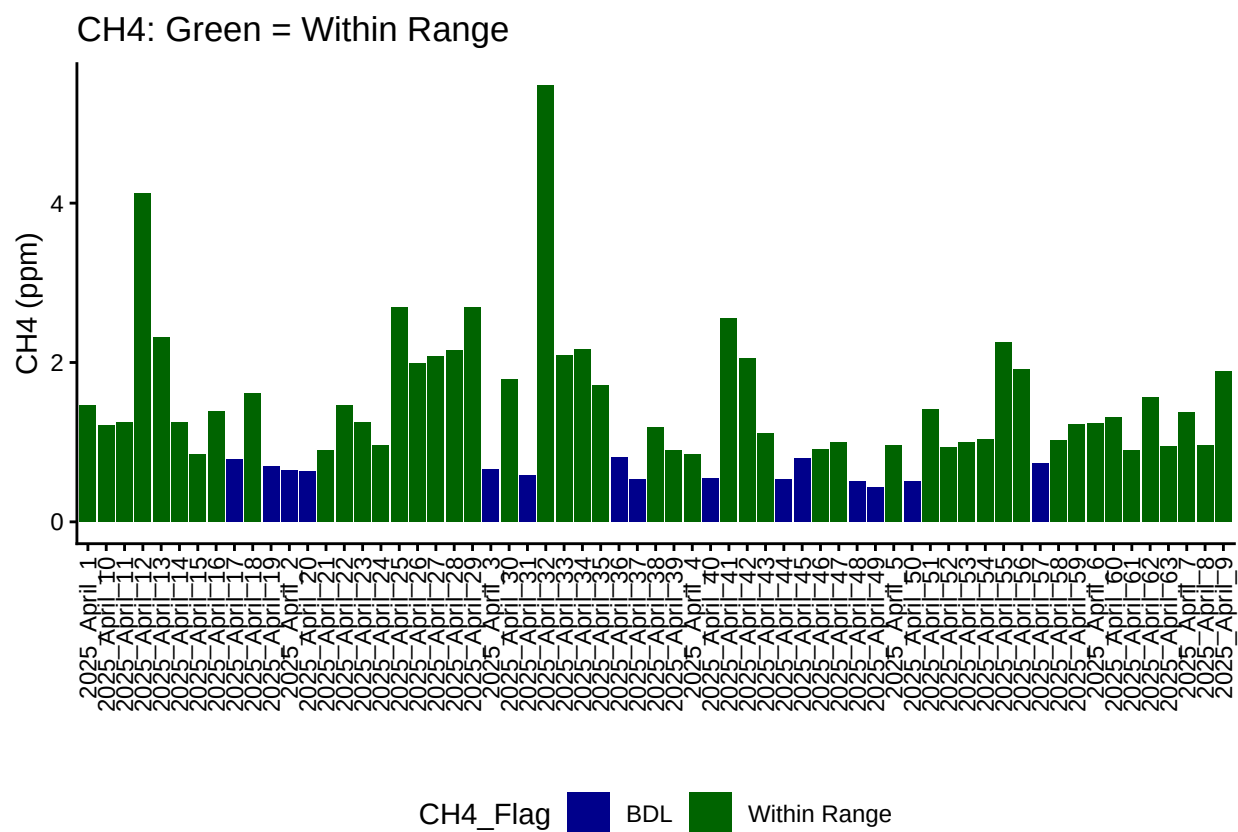


Blank Conc <25% Quartile Samples ■ YES

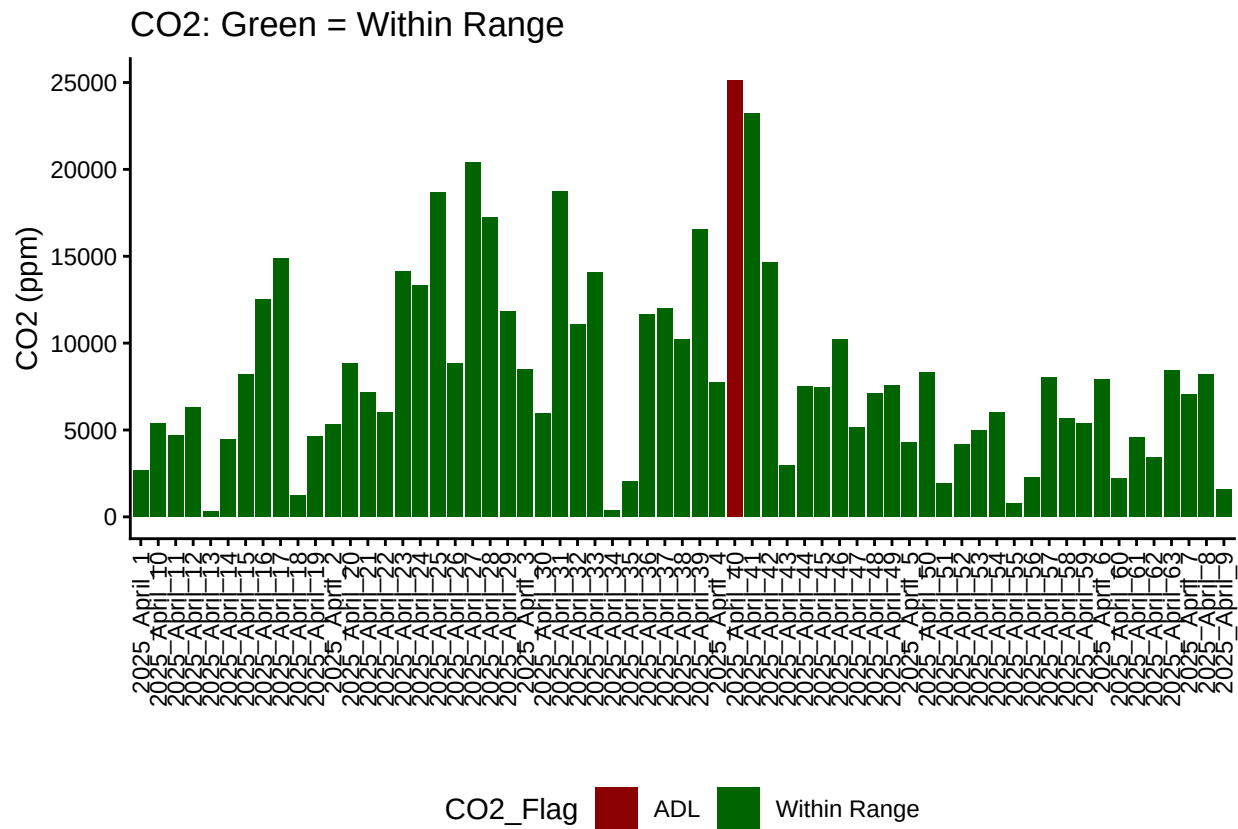
Table 3: Mean Area of Blanks

Test	Blank_Mean_Area
N2O	0

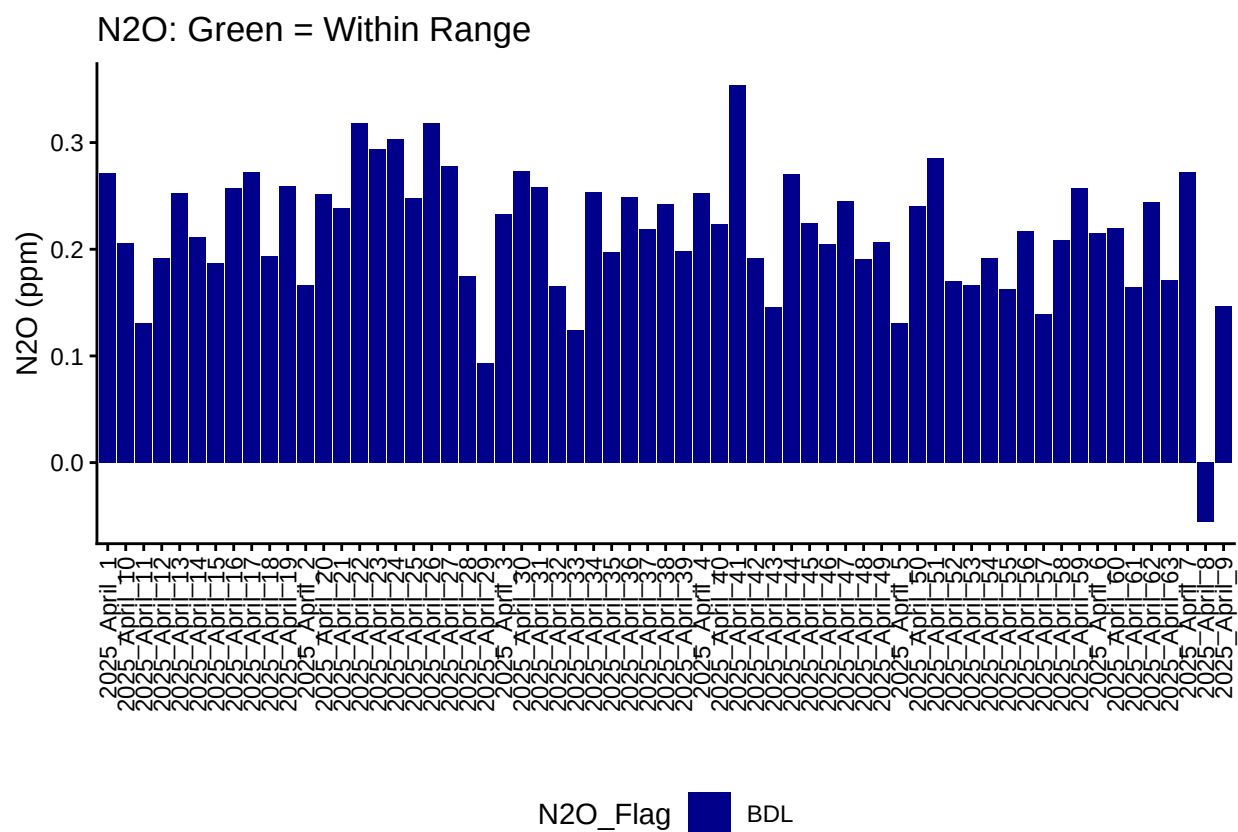
10 Check Range - CH4



11 Check Range - CO2

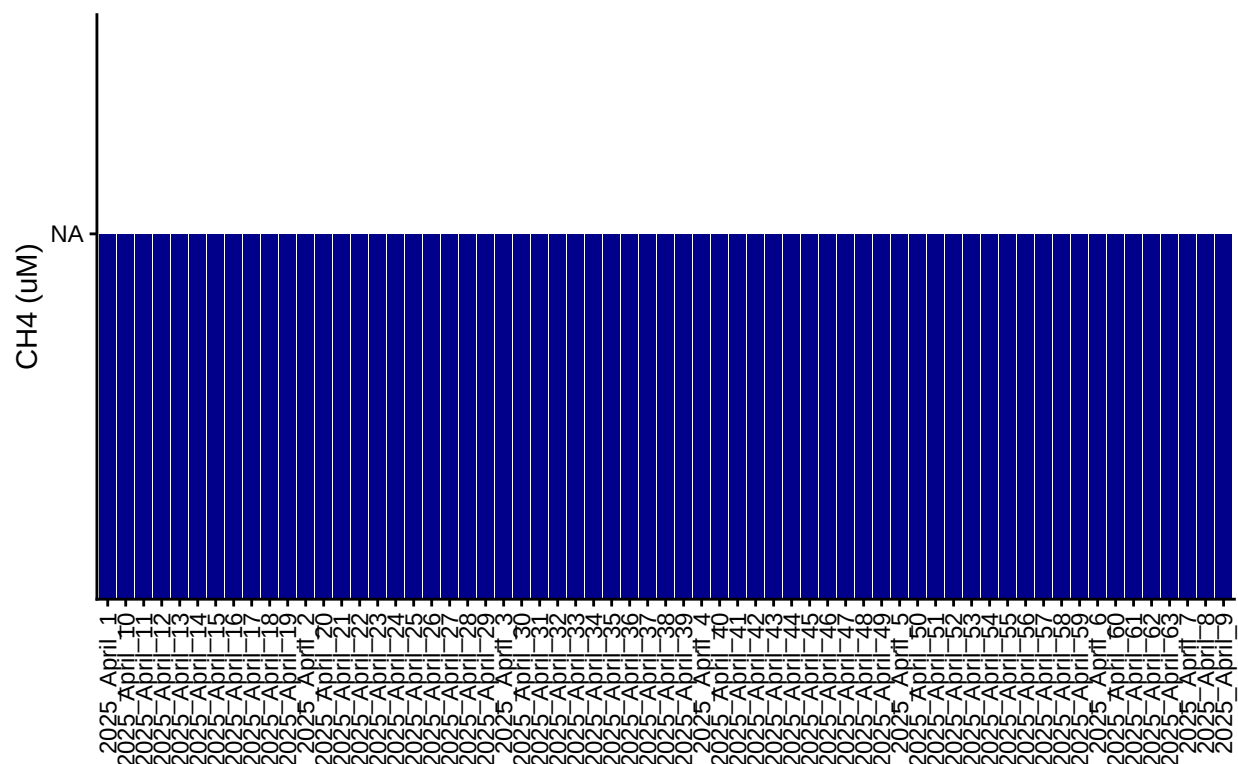


12 Check Range - N2O

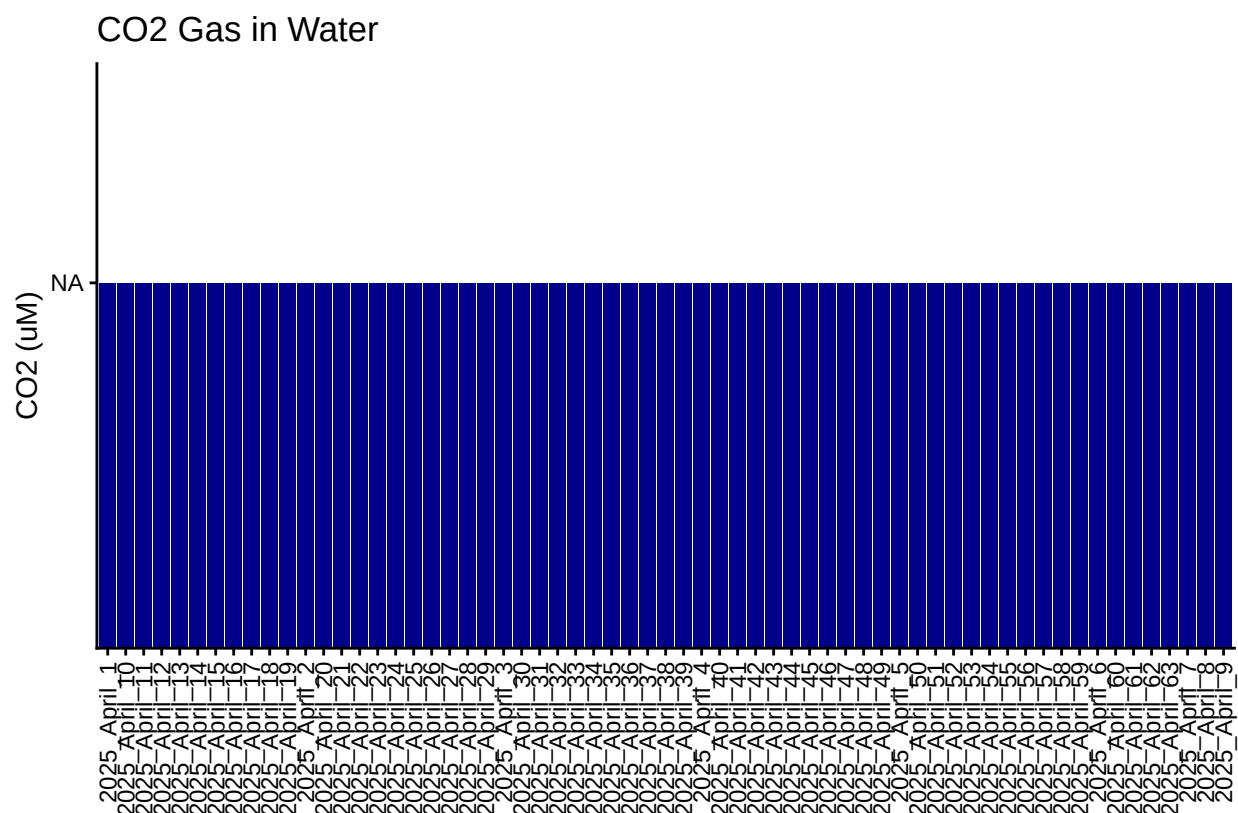


13 Calculate gas in water - CH₄

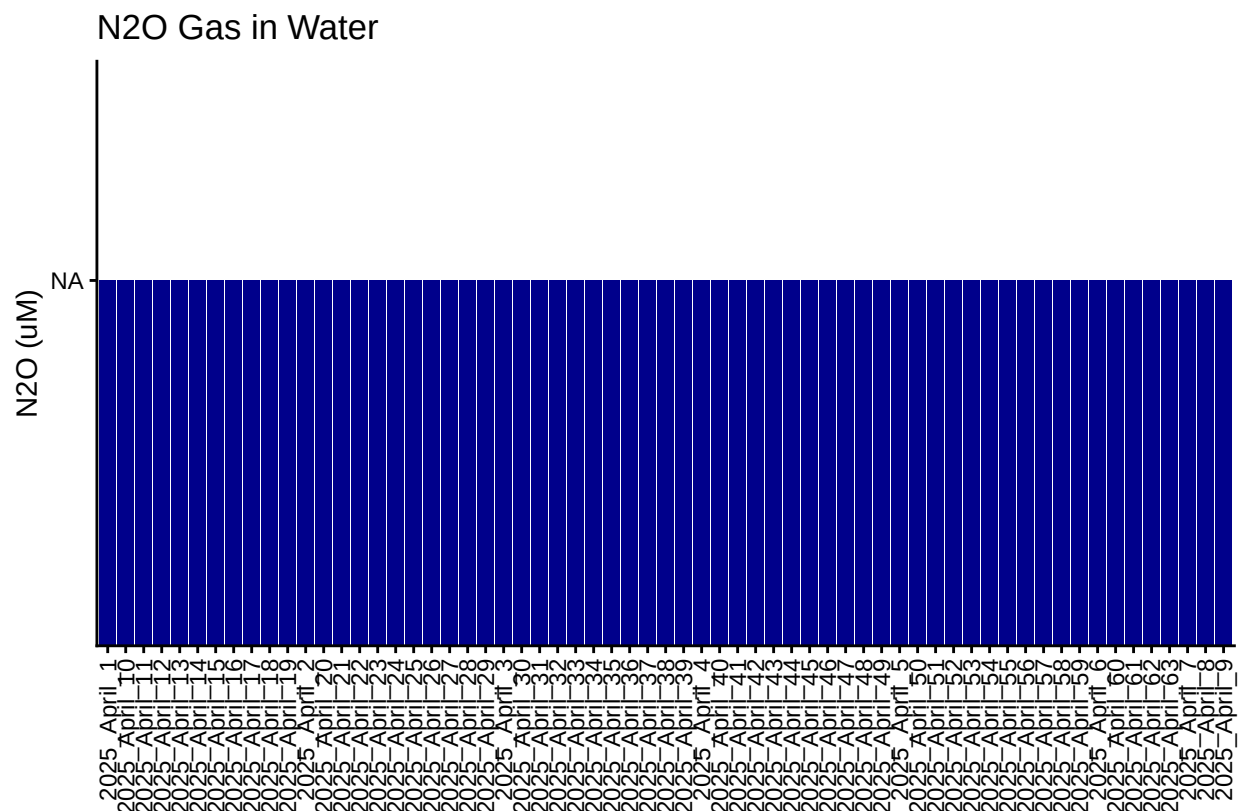
CH₄ Gas in Water



14 Calculate gas in water - CO₂



15 Calculate gas in water - N2O



16 Merge samples with Metadata & check all samples are present

Merge Samples with Metadata

```
## All sample IDs are present in data
```

16.1 Organize Data

##	Sample_ID	Type	Run_Date	CH4_Area	Conc_CH4	CH4_RTime_Flag	C02_Area
## 1	2025_April_1	Sample	2026-08-26	736	NA	Correct - CH4 RTime	6701
## 2	2025_April_2	Sample	2026-08-26	504	NA	Correct - CH4 RTime	13227
## 3	2025_April_3	Sample	2026-08-26	510	NA	Correct - CH4 RTime	21113
## 4	2025_April_4	Sample	2026-08-26	563	NA	Correct - CH4 RTime	19235
## 5	2025_April_5	Sample	2026-08-26	594	NA	Correct - CH4 RTime	10635
## 6	2025_April_6	Sample	2026-08-26	674	NA	Correct - CH4 RTime	19585
##	Conc_C02	C02_RTime_Flag	N20_Area	Conc_N20	N20_RTime_Flag		
## 1	NA	Correct - C02 RTime	580	NA	Correct - N20 RTime		
## 2	NA	Correct - C02 RTime	427	NA	Correct - N20 RTime		
## 3	NA	Correct - C02 RTime	524	NA	Correct - N20 RTime		
## 4	NA	Correct - C02 RTime	553	NA	Correct - N20 RTime		
## 5	NA	Correct - C02 RTime	374	NA	Correct - N20 RTime		

```

## 6      NA Correct - CO2 RTime      498      NA Correct - N2O RTime
##  FID_Peak_Flag ECD_Peak_Flag CH4_Conc_ppm      CH4_Flag CO2_Conc_ppm
## 1      Correct      Correct      1.4568356 Within Range      2670.549
## 2      Correct      Correct      0.6416444      BDL      5317.775
## 3      Correct      Correct      0.6627269      BDL      8516.675
## 4      Correct      Correct      0.8489560 Within Range      7754.877
## 5      Correct      Correct      0.9578824 Within Range      4266.348
## 6      Correct      Correct      1.2389828 Within Range      7896.852
##      CO2_Flag N2O_Conc_ppm N2O_Flag CH4_Conc_uM CO2_Conc_uM N2O_Conc_uM
## 1 Within Range      0.2714449      BDL      NA      NA      NA
## 2 Within Range      0.1667880      BDL      NA      NA      NA
## 3 Within Range      0.2331391      BDL      NA      NA      NA
## 4 Within Range      0.2529760      BDL      NA      NA      NA
## 5 Within Range      0.1305342      BDL      NA      NA      NA
## 6 Within Range      0.2153543      BDL      NA      NA      NA

##  Sample1 Sample_Year Sample_Month Sample_Day Sample_Time Time_Zone G_W
## 1      SGW      2025      April      16      1054      EDT      G
## 2      SGW      2025      April      16      1055      EDT      G
## 3      SGW      2025      April      16      1056      EDT      G
## 4      SGW      2025      April      16      1057      EDT      G
## 5      SGW      2025      April      16      1059      EDT      G
## 6      SGW      2025      April      16      1100      EDT      G
##  Sample_Plot Sample_Grid Sample_Depth Field_Notes Lab_Notes CH4_Conc_ppm
## 1      Control      C3      10      NA      NA      1.4568356
## 2      Control      C3      20      NA      NA      0.6416444
## 3      Control      C3      35      NA      NA      0.6627269
## 4      Control      C3      50      NA      NA      0.8489560
## 5      Control      C6      10      NA      NA      0.9578824
## 6      Control      C6      20      NA      NA      1.2389828
##  CH4_Conc_uM CH4_Conc_Flag CO2_Conc_ppm CO2_Conc_uM CO2_Conc_Flag N2O_Conc_ppm
## 1      NA      Within Range      2670.549      NA      Within Range      0.2714449
## 2      NA      BDL      5317.775      NA      Within Range      0.1667880
## 3      NA      BDL      8516.675      NA      Within Range      0.2331391
## 4      NA      Within Range      7754.877      NA      Within Range      0.2529760
## 5      NA      Within Range      4266.348      NA      Within Range      0.1305342
## 6      NA      Within Range      7896.852      NA      Within Range      0.2153543
##  N2O_Conc_uM N2O_Conc_Flag      QAQC_Flags Analysis_rundate Run_notes
## 1      NA      BDL N2O_R2 is Bad      2026-08-26
## 2      NA      BDL N2O_R2 is Bad      2026-08-26
## 3      NA      BDL N2O_R2 is Bad      2026-08-26
## 4      NA      BDL N2O_R2 is Bad      2026-08-26
## 5      NA      BDL N2O_R2 is Bad      2026-08-26
## 6      NA      BDL N2O_R2 is Bad      2026-08-26

```

16.2 Visualize Data CH4

```
## Visualize Data CH4
```

TEMPEST: Soil Gas Well CH4



16.3 Visualize Data CO2

```
## Visualize Data CO2
```

TEMPEST: Soil Gas Well CO2



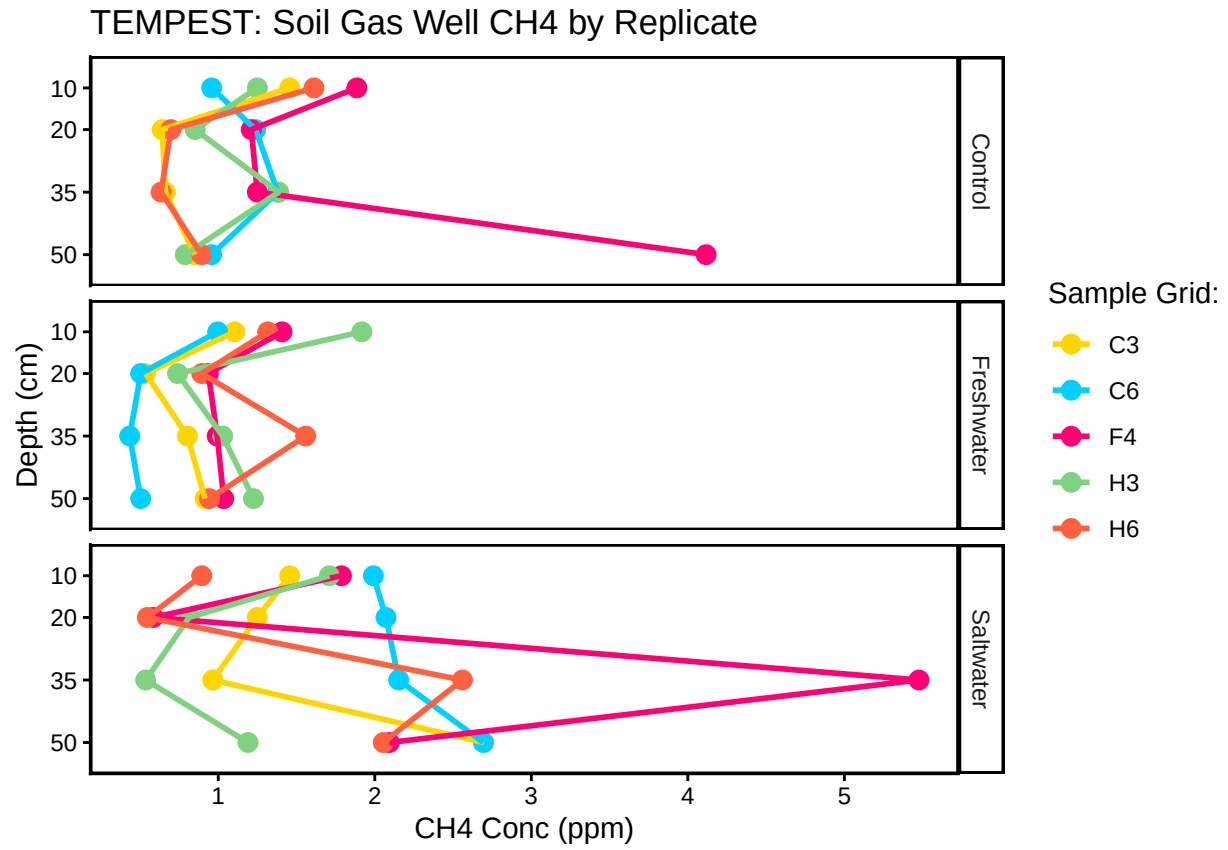
16.4 Visualize Data N2O

```
## Visualize Data N2O
```

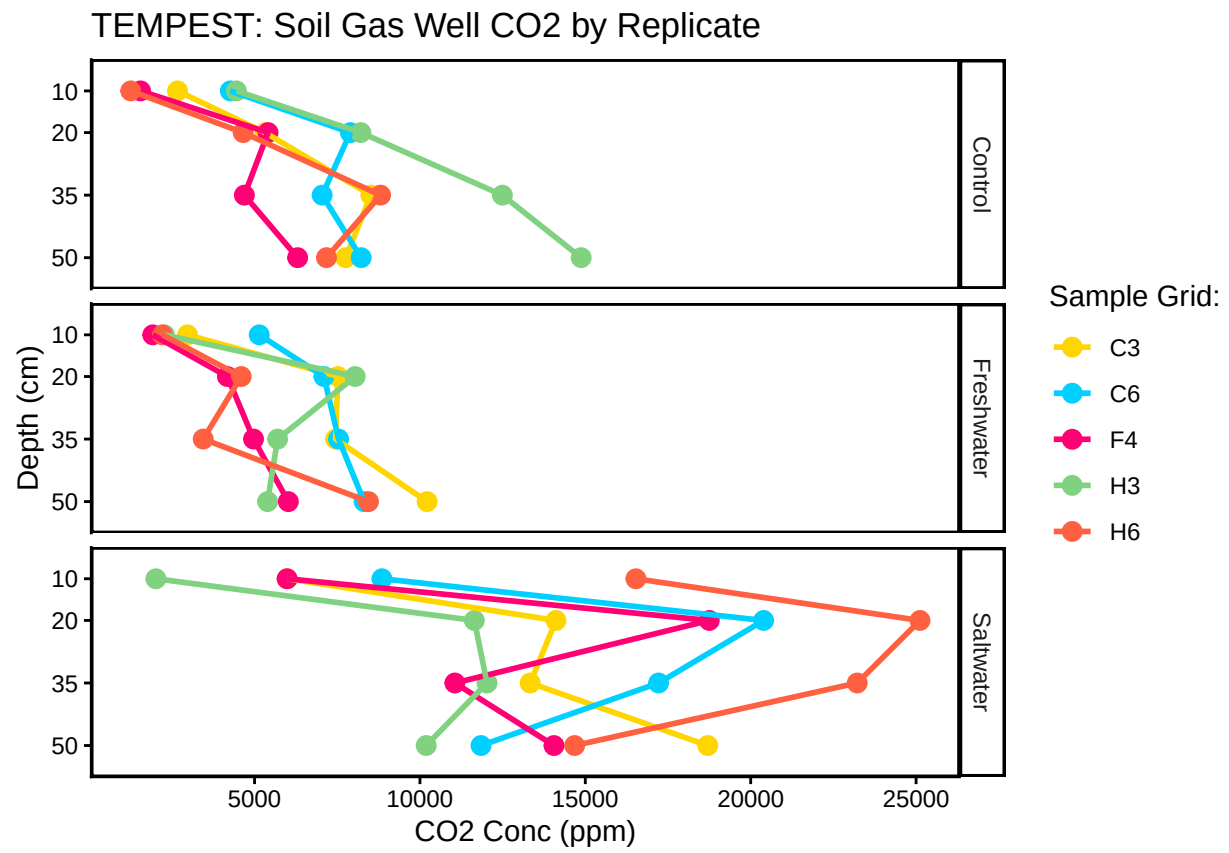
TEMPEST: Soil Gas Well N2O



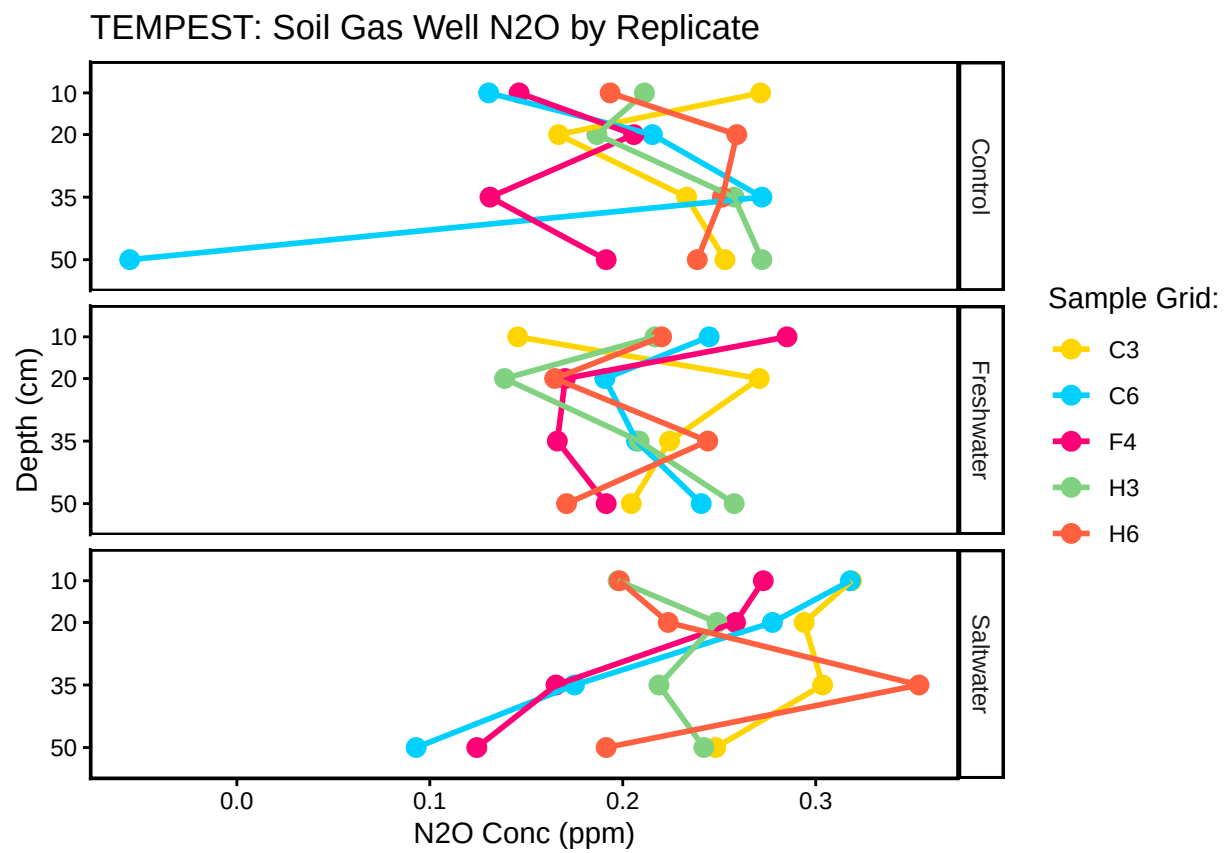
#Visualize Data CH4 by Replicate



#Visualize Data CO2 by Replicate



```
#Visualize Data N2O by Replicate
```



```
##Export Data
```

```
#end
```